

A Benchmark asset construction

Input sources

HERB-derived herb-target associations

HERB-derived herb-compound context

ChEMBL-derived compound-target context

Target-target context

Identifier harmonization and quality control

- 1 Herb identity normalization
- 2 Target harmonization (Unified Medical Language System)
- 3 Duplication / ambiguity screening
- 4 Benchmark eligibility filters

Frozen benchmark assets (protocol v1.0)

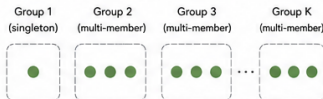
Herbs	1,247
Targets	2,309
Corrected herb-target positives	87,825
Formal eligible herb-target positives after quarantine	86,317
Herb-compound edges	14,985
High-strict compound-target edges	1,456
Target-target unordered pairs	2,536

H0847 quarantined

1,508 positive pairs excluded from all splits and frozen as excluded in protocol v1.0

B Split-blocking and leakage-control

Split-blocking groups (1,051 total)



Group-aware split (no group crosses partitions)

Train	Validation	Test
535 herbs	67 herbs	67 herbs
339 groups	67 groups	67 groups
69,053 positives	8,632 positives	8,632 positives



Split-blocking statistics

Total split-blocking groups	1,051
Singleton groups	941
Multi-member groups	110
Largest group size	9

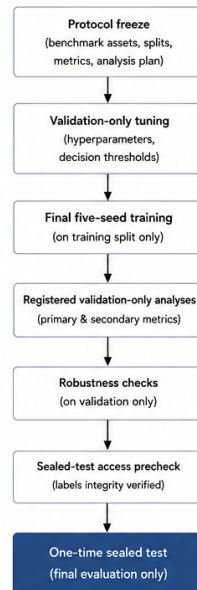
Context-only herbs excluded (not in any split): 577

Leakage-control principles

- Held-out herbs excluded from training graph.
- No split-blocking group crosses partitions.
- Operational split blocking controls exact normalized-name identity route only; it is not a canonical medicinal-material equivalence claim.

C Frozen evaluation workflow

Workflow (frozen after protocol v1.0)



Audit guardrails (enforced)

Test labels unseen before final test.

No post-test tuning.

No post-test model selection.

No post-test design change.

One-time sealed test (summary)

Test herbs	67
Recorded positive pairs	8,632
Candidate targets	2,309